

International Conference on Industry 4.0 and Smart Manufacturing (ISM 2019)

Exoskeletons Towards Industrie 4.0: Benefits and Challenges of the IoT Communication Architecture

Enrique Bances^{a,b,*}, Urs Schneider^{a,b}, Jörg Siegert^a, Thomas Bauernhansl^{a,b}^a*Institute of Industrial Manufacturing and Management, University of Stuttgart, Nobelstr. 12, 70569 Stuttgart, Germany*^b*Fraunhofer Institute for Manufacturing Engineering and Automation IPA, Nobelstr. 12, 70569 Stuttgart, Germany*

Abstract

The volatility and flexibility of a production line in a smart factory demands for high dynamics during the assembly tasks and repetitive movements in an un-ergonomic body posture. Hence, the workers are exposed to the danger of developing work-related musculoskeletal disorders (MSDs). Previous studies have reported that exoskeletons help to reduce fatigue and prevent injuries during lifting, carrying, pushing, pulling, and overhead-positions tasks, improving productivity in manufacturing industry. However, no studies have reported the advantages of digital integration of an exoskeletons network in a smart factory. This paper gives an overview of benefits and challenges regarding the connectivity of exoskeletons within a smart factory. Moreover, it describes the communication architecture of a cyber-physical exoskeleton network. Finally, an experimental test of fog computing structure was conducted based on IoT messaging protocols, allowing real-time data analytics for benefits of the manufacturing process.

© 2020 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the International Conference on Industry 4.0 and Smart Manufacturing.

Keywords: Exoskeleton; Industry 4.0; CPS; IoT; Fog/Cloud computing; Big Data & analytics

1. Introduction

The growing customers demand for products suiting their individual requirements (mass customization), the global market competition, and the complexity of the production chain to satisfy those demands are some challenges that a modern manufacturing system must face. As a consequence, manufacturers require automation tools that can rapidly adapt to the changing needs, without excessive downtime or expenses. This also should not affect their production run or increase manufacturing costs. Based on this a smart factory (SF) is capable of managing this complexity [20].

SF is based on cyber-physical systems (CPS) and an intelligent network of machines and products. The product itself communicates the information required by the SF for manufacturing. According to this information, the individual production steps are controlled to obtain the desired final result.

The major features of an SF are connectivity, optimization, transparency, pro-activity, and agility. They enable well-funded decisions and help to improve the production process [10, 24, 31]. Moreover, to achieve the connectivity in the sector of industrial automation and IoT; OPC-UA, MQTT, DDS-I and ROS are the most commonly use communication protocols and frameworks [25].

On the other hand in the smart factories workers are still exposed to various production activities such as assembling, physical work-loads (material handling), repetitive movements such as lifting and moving parts or products into the assembly line, frequent bending and un-ergonomic body posture. Hence, they are exposed to develop a work-related musculoskeletal disorder (MSD). In these cases, an exoskeleton represents a feasible solution as an ergonomic tool in order to reduce MSD in manual manufacturing.

An extensive overview of exoskeletons on the market and the developed prototypes at universities or research institutions shows that exoskeletons can be classified into two categories: (1) a passive system does not use any type of actuator, but rather use materials, dampers and springs to support a posture or motion. (2) An active exoskeleton comprises actuator to augments the human power and it supports the joints during the operation [11, 30].

* Corresponding author. Tel.: +49-711-970-3711 ; fax: +49-711-685-51892.

E-mail address: enrique.bances@iff.uni-stuttgart.de (Thomas Bauernhansl).

In the last five years, the interest for using passive exoskeleton in assembly lines has risen. For instance, Ford Motor Company has tested the EksoVest (designed by a partnership between Ford© and Ekso Bionics®) and recorded up to 83% decrease of injuries at their assembly line during the years 2015-2016 [6]. Also, a study conducted by Iowa state university which employed Electromyography (EMG) sensors attached to workers to vet the Airframe (designed by Levitate Technology, Inc) concluded that fatigue reduced 18% to 39% in the shoulders and biceps [21]. However, there are no reported studies that show exoskeletons using IoT technologies and a communication architecture for the integration of these devices into a SF.

The purpose of this paper is to describe the implementation of an exoskeleton's network based on IoT communication architecture, fog computing (data analytics), and the development of cloud services. The paper is organized as follows: In Section II, a brief resume on the two main common issues during manual assembly tasks such as the development of MSD and the cognitive factor is presented. In Section III, the key IoT technologies involving an exoskeletons network and trends in communication, fog/cloud computing to develop stream processing and data analytics are given. Section IV describes the implementation of an exoskeleton's network based on MQTT communication protocols. The method used for stream processing and data analytics, as well as the developments of software applications and services on the cloud is described. In Section V, main issues and challenges of the key technologies are presented. Finally, in Section VI a brief conclusion and vision of further development of exoskeletons integration into a SF are given.

2. MANUAL ASSEMBLY LINE ISSUES IN I4.0

Despite the high connectivity and automation in a SF, the manual assembly is still a fundamental part of the production line. The continuous varying of products and tasks demand a high level of flexibility and cognitive abilities. It means, human beings' capacities to observe, decide, and adopt proper actions are required. However, the factory workers are still exposed to physical workload, repetitive movements, frequent bending and un-ergonomic body postures. Hence, they are exposed to the danger of developing work-related musculoskeletal disorders (MSD).

On the other hand, due to product variation and complexity in manufacturing, the assembly operator needs well-designed and appropriate information at the right time to perform the correct, effective and accurate task. Otherwise, there is a high risk that this variation affects the time and increases the errors during the assembly [8, 9].

2.1. Development of MSD

MSDs are one of the most common work-related complaints, affecting millions of workers and costing billions of euros to the companies. As for factory workers, these MSDs can make their

working life painful and in many cases bring it to early retirement. There are several studies that reported how the MSDs affect the health of the workers. For instance, in 2015, more than 61% of workers reported any kind of MSD in Europe [14], increasing by 21% compared to the previous study carried out in 2012 [3].

Particularly in automotive industry, a high discomfort and ergonomics problems that affect workers health were notified [23]. Other studies affirmed that about 70% of job tasks have a very high risk of MSDs for neck posture and 60% for the back (in moving condition) and shoulder/arm posture [4].

In an automotive industry, some solutions have been implemented for reducing MSDs e.g., the usage of a cantilevered seat for some interior assembly process of the vehicle. The effectiveness of the chair depends on the region of the vehicle the assembler works on. The experimental results show that in most cases, the spine loads were reduced but not the shoulder muscle activity [11]. Furthermore, this solution has the disadvantage that it requires a certain time to fit the chair in the correct and comfortable position for working. However, other ergonomics solutions that can reduce MSDs are necessary for the following activities:

- The "overhead-positions" tasks which conduce to a high probability of shoulder/arm pain
- The tires installation which conduce to a wrist, shoulder and arm pain
- The auto seat installation which affects the muscles and joints in the shoulders, the arms, back, and the wrists
- Logistic tasks such as lifting and carrying heavy components packed in boxes on hip height
- Logistic tasks such as picking and putting objects from/into a pallet cage

2.2. Cognitive Factor

In dynamic manufacturing (modern factories), the continuous varying of products and tasks, demand a high level of flexibility and cognitive abilities, where still human capacity to observe, decide, and adopt proper actions are required.

These variations of tasks can affect the assembly process and there is a high risk of assembly errors. The worker is likely to show an automatic behavior after a series of the same product and mix information from one product with another. Consequently, productivity and even product quality are affected. Therefore, in a manual assembly line, it is important that the worker receives the correct information at the right time to perform an effective and accurate assembly tasks. The issue of this system is that the workers need to be prepared for different configuration types with different components of each product and this puts considerable strain on them [8].

Some experimental studies have shown that the products variants increase the complexity in manufacturing and have a negative effect on productivity and product quality. The reason is the large number of components to be handled, as well as an insufficient presentation of the supplies and assembly information in the workstation [9].

3. IoT Technologies and Trends Involving Exoskeleton Network

3.1. Exoskeleton as a Cyber-physical System

The CPS are entities which are in intensive connection with the surrounding physical world providing and using, at the same time, data-accessing and data-processing services available on the Internet (Monostori 2016). Based on this definition, an active exoskeleton needs to change from being a traditional system built upon an embedded system to be a CPS as part of a network of interacting elements through the wireless communication and its connection to the Internet (cloud). In case of a passive exoskeleton, a group of sensors and wireless communication module should be installed. Thus, the CPS sensors allow monitoring the working condition of the exoskeletons. The results of the processed data provide information for improving the performance of a production line, integrating of the supply chain, connecting the shop floor to the board-room creating highly flexible production, and creating substantial cost savings in a SF. This interconnection of CPS through the internet in a SF with regard to safety, reliability and security is called Industrial IoT (IIoT).

3.2. IoT Messaging Pattern and Protocols

The key of messaging patterns is to understand how messages are transported in the network to accomplish a certain task. Two of the commonly used patterns are Request/Response and Publish/Subscribe.

- Request/Response (req/resp): It is a basic communication pattern. It consist of a client that request information from a server. Previously a linkage is stablished between them
- Publish/Subscribe (pub/sub): It allows mass distribution of information from one publisher to multiple interested subscribers reducing the network traffic. The publisher does not have to send individual messages, it uses an intermediary node (broker) to distribute the information

Regarding IoT protocols, there are publish/subscribe, request/response or both patterns are commonly used in standards. The objective of this paper is not to delve into detail about the characteristics of each protocol. However, a brief overview of the most commonly implemented IoT protocols relevant to this study are given. For instance, MQTT (Message Queuing Telemetry Transport) is a lightweight IoT protocol, which uses pub/sub communication pattern, is used for machine-to-machine (M2M) communication. It is specially designed to work in low bandwidth scenario and it ensures efficient communication while consuming the least possible battery on hardware devices [22].

Further, strengthening the concept of Industrial IoT (IIoT), OPC-UA is an industrial automation protocol that initially used only a req/resp architecture as a communication pattern, However, the pub/sub extension was subsequently released in 2018[12], enabling the implementation of universal data inter-

connectivity and information interoperability. The pub/sub flavor of this protocol integrates characteristics of existing protocols MQTT and AMQP, such as defining a broker for a message distribution and enable the use of this protocol over the internet, but retaining the end-to-end security and standardized data modeling advantages. Besides, OPC-UA pub/sub uses the UDP transport protocol for establishing low-latency and it was listed as the communication protocol in the reference architecture model for Industrie 4.0 (RAM I4.0) [1].

3.3. Communication Architecture for CPS

There are three scenarios for futures solution regarding the automation architecture of any CPS and its integration with a cloud platform: Evolution, Progression, and Revolution [5].

The evolution architecture consists of interconnecting a system using a bridge or smart connector to a manufacturing platform. For example, a robot with installed communication protocol such as OPC-UA or MQTT could be connected to an IoT-gateway. This Gateway with an appropriate configuration and an installed platform interface allows the connection of the robot and the fog/cloud-platform.

The progression architecture is based on a modular software architecture (MSA) installed in the machine. The MSA refers to the design and high-level communication between components (modules) on different levels of the control architecture of any system. Thus, it allows replacing or adding any module without affecting the rest of the system. There are several frameworks for robotics systems which allow for a modular, scalable robotic software architecture e.g. ROS, HoP, Aria, Orocos, and Orca [29]. However, ROS is the most commonly used open source framework in the research community.

3.4. Data Analytics and Stream Processing

The science of examining raw data with the purpose of drawing conclusions about that information is called data analytics. Data analytics techniques can reveal trends and metrics that would otherwise be lost in the mass of information. This information can then be used to optimize processes to increase the overall efficiency of a system. The analytics process starts with data collection (e.g., sensor data ingestion), in which the information is identified for a particular analytics application. Then it involves applying an algorithm (stream processing) to derive insights [32].

Stream processing also called real-time analytics is a technology that lets users query continuous data streams and detect conditions quickly within a short period from the time of receiving the data. The detection time period just take few milliseconds [18]. For example, we can query data streams coming from the exoskeleton position sensors located on the forearms and receive an alert when the ergonomics position during the tasks is not adequate. As a consequence, it could prevent future joint or muscular injuries.

The last step in the data analytics process is communicating the results to the users to support their decision-making (responsible persons, software services, robots, or machines). This

usually is done with the help of data visualization techniques, data visualizations often are incorporated into dashboard applications that display data on a single screen and can be updated in real time as new information becomes available.

3.5. Fog/Cloud Computing in an Exoskeleton Network

Fog computing allows processing data locally, rather than send all of this data to the cloud. It reduces the possibility of connection loss, due to the multiple interconnected channel. Also, fog computing increases the network efficiency permitting data to be processed and accessed more rapidly and efficiently, which reduces the risk of data latency. On the other hand, cloud computing has the role in processing a large amount of data coming from IoT devices. However, the long distances between the clients and the data processing center increasing the transmission latency [2, 7].

Fig. 1 shows an example of communication architecture for exoskeletons network. The fog layer acts like a mediator between data centers in the cloud over Internet and the local IoT devices and CPS as the exoskeletons. In this layer, the data center (exo-server) allows ingesting of data from the multiple devices and the use of platform for stream processing. Also, the data center offers the development of several digital solutions such as services and application. For instance, the data center allows the long term storage of data, the visualization and monitoring of sensors signals, as well as running machine learning algorithms to improve the performance of the exoskeleton. The high layer of this architecture connect the exoskeleton network with the cloud via Internet, allowing to access relevant information from another sector e.g., production planning, product development instruction or product assembly characteristics.

4. Implementation of Communication Architecture for Exoskeleton Networks

As an example, the developed method shows how an active exoskeleton was integrated into a pub/sub topology network. Besides, the network is connected to a data center called exo-server. This server works as such a broker and allows developing stream data analytics. The experiment used an active upper-body exoskeleton "Stuttgart Exo-Jacket 2.0" (SEJ2.0) developed by Fraunhofer IPA Stuttgart, Germany. SEJ2.0 is a personalized system developed to attend carrying and lifting functions (Fig. 2b). It counts with two brushless motors in each arm (shoulder and elbow). Each motor has an encoder sensor which allows measuring the position, velocity acceleration of the movement [13]. Also, each arm has two strain gauge sensor used to detect the movement intention and the development of the control algorithm (See Fig. 2a). The exoskeleton integrates a microcontroller-based board (sensor board) in each arm. This board includes a WiFi module (esp8266), allowing the sensor data transmission.

The implemented communication was MQTT pub/sub pattern-based. Finally, the data transmission and communication code were written and uploaded via Arduino IDE. The fog

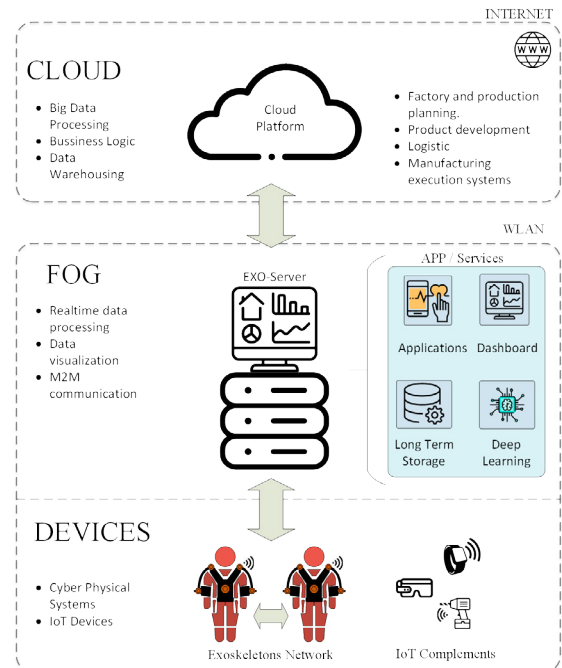


Fig. 1. Communication Architecture of exoskeletons network. It shows three levels of bidirectional data transmission. The lower level permits the connection of exoskeletons and others CPS based on IoT protocols. The second level allows ingesting sensors data and real time stream data processing (fog computing). In this level other technologies are applied such as deep learning algorithms, long term storage in databases and software application for interacting with the network devices. The highest level is the cloud computing. On this level big data analysis is developed, accessing relevant information from another sector in the factory (e.g., production planning, product development instruction or product assembly characteristics).

computing is implemented in a server based on Ubuntu Server 18.04.2 LTS. The raw data collection is conducted for Kafka¹, an open-source message-broker software. To process real-time data, Flink² Streaming was used. Flink enables the implementation of complex event processing algorithms which allows for pattern detection in event streams. The goal of this algorithm is to send alerts when an unusual performance of the exoskeleton during the task happened. Two cases were tested with the algorithm. (1) The detection of an unusual repeated increase of current of the motors (elbow and shoulder) in a short period of time during the task. (2) The detection of values that exceed the allowed ranges of movement during the use of the exoskeleton. In this example, the angular velocity and the angle of each motor are monitored. The objective is to prevent some future breakdowns in the engines and the mechanical structure of the exoskeleton. However, other applications are considered to implement in future works such as monitoring the exoskeleton's performance, ergonomic features during the activities, efficiency and health indicators of the user, as well as to calculate

¹ Kafka® is used for building real-time data pipelines and streaming apps.

² Apache Flink allows running real-time data processing pipelines in a fault-tolerant way at a scale of millions of events per second.

and evaluate productivity ratios of the assembly process. Then, this information can be shared with other CPSs or persons using application programming interfaces (APIs) or web portals.

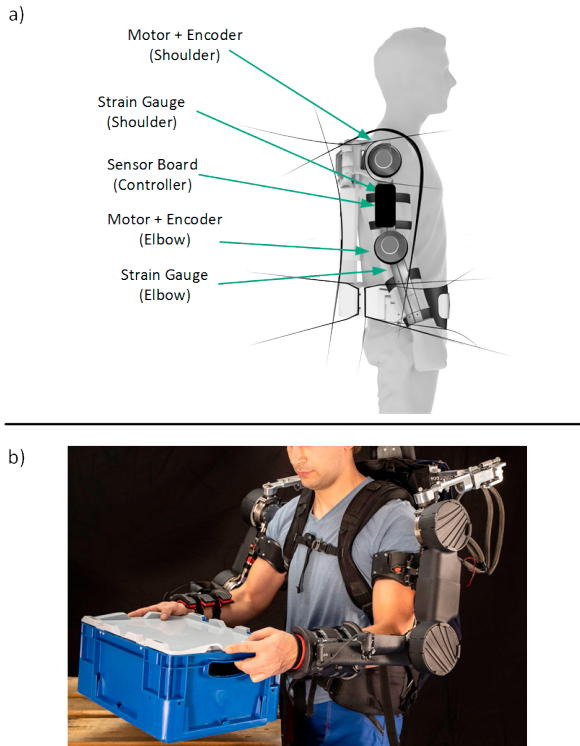


Fig. 2. a) SEJ2.0 electronics architecture: sensors and actuators distribution, b) Implemented SEJ2.0 during lifting and carrying tasks.

Finally, the processed data take a specific direction depending on the application. For instance, the data return to the exoskeleton or are sent to other devices as informative or alert messages. The data is stored in a long term database (Influxdb³) and visualized in a dashboard (Grafana⁴). Fig. 3 shows the implemented data analytics model on the server.

On the other hand, the interaction with the exoskeleton network is conducted by a developed web platform (SEJ2.0 APP). This app works as a publisher and subscriber and exchanges information with all the components in the network. It sends personalized parameters such as the maximal weight to carry to the exoskeleton allowing setting up the performance of the system before or during activity. Also, it receives alerts from the data center the results of processed data. SEJ2.0 APP was tested with a tablet, smartphone, and desktops.

In addition to the exoskeleton, another seven devices were connected to the network. Two smartphones based on Android

OS (subscribers), four extra sensor boards emulating sensor data (publisher), and one desktop PC (publisher) with Ethernet connection. The smartphone and the desktop used MQTT client applications to receive and send messages.

The whole experiment was conducted in the laboratory of Biomechatronic Systems Group of Fraunhofer-IPA. The experiment consisted of lifting and carrying boxes (1kg, 3kg, and 5kg respectively) located on a standard dimensions table and displace them from the table to a 1.5 meters high shelf.

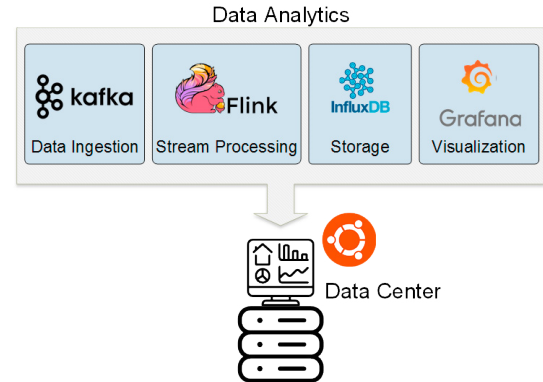


Fig. 3. SEJ2.0 data center. The server deploys big data & analytics models (Fog computing).

5. Results and Future Work

The following observations were made as a result of experiments conducted with an exoskeleton within a publish/subscribe communication architecture. One subscriber could receive information from many different publishers and one publisher could send data to many different subscribers. This allowed the rapid exchange of a large volume of data between many devices in the network at the same time. Also, it reduced network traffic and latency because publishers do not need to know about the number of subscribers, what topics a subscriber is listening to or how subscribers work. However, the broker does not notify the subscribers and publishers about the message status (failed or successful deliveries). Although in this example only seven devices and an active exoskeleton were part of the network, this architecture presented the feature of scalability. In theory, an infinite number of devices could connect to the network. However, the increasing number of messages being exchanged could lead to instabilities in this architecture. The used broker (server) provided satisfactory results with six connected devices. The experiment reported good results of connectivity using MQTT. However, a future test using OPC-UA protocol or ROS framework as high communication level will give advantages to integrate the exoskeleton directly in the OPC-UA ecosystem in the factory. During the experiment time (approx. 2 hrs.) disconnections were not recorded from the publishers. However, the subscribers presented sudden disconnection and re-connection. The reason was that both the laptop

³ Influxdb is an open-source time series database, optimized for fast, high-availability storage and retrieval of time series data in fields such as operations monitoring, application metrics, IoT sensor data, and real-time analytics.

⁴ Grafana is an open source metric analytics & visualization suite. It is used for visualizing time series data for infrastructure and application analytics

communication pattern and the IoT protocol MQTT. Also, we showed that exoskeletons can be connected to a server platform for storing and processing sensors data to share useful information through the Internet, in order to increase productivity, prevent future injuries in the workers and reduce error in the assembly process.

Fig. 4 shows an example of exoskeleton working in a smart automotive factory. Despite the high connectivity and automation in a smart factory, manual assembly is still a fundamental part of the production line. Also, considering the product variation, and complexity in manufacturing, the assembly operator needs the appropriate information to perform the correct, effective and accurate task. This system can share information about the productivity of the operators during the task, allowing better and anticipated planning. Also, exoskeletons share information about movements, positions, and applied forces during the activities, reducing the risk of developing injuries. Moreover, exoskeletons can interact with other systems such as automated guided robots in the supply chain, assembly tools, automated machines, and autonomous software, saving time during the assembly process of different products and reducing operators' mistakes. Finally, this paper has described the exoskeletons as powerful communication tools in the future manufacturing process.

Acknowledgements

This article was written with the collaborative research of the Fraunhofer Institute - IPA and the IFF of University of Stuttgart. The research is part of the project "Leistungszentrum" Mass Personalization Stuttgart (Pilot I - Personalized Biomechatronic Assistance Systems).

References

- [1] Adolphs, Peter, et al., 2015. Status Report-Reference Architecture Model Industrie 4.0 (RAMI4.0). VDI-Verein Deutscher Ingenieure eV and ZVEI-German Electrical and Electronic Manufacturers Association, Tech. Rep.
- [2] Aazam, M., Zeadally, S., & Harras, K. A., 2018. Deploying fog computing in industrial internet of things and industry 4.0. *IEEE Transactions on Industrial Informatics*, 14(10), 4674–4682.
- [3] Anxo, D., Franz, C., & Kümmerling, A., 2013. Working time and work–life balance in a life course perspective: a report based on the fifth European Working Conditions Survey. Eurofound.
- [4] Aziz, F. A., Ghazalli, Z., Mohamed, N. M. Z., & Isfar, A., 2017. Investigation on musculoskeletal discomfort and ergonomics risk factors among production team members at an automotive component assembly plant. In *IOP conference series: materials science and engineering*, 257(1), 012040.
- [5] Bauernhansl, T., Ten Hompel, M., & Vogel-Heuser, B., 2014. *Industrie 4.0 in Produktion, Automatisierung und Logistik: Anwendung-Technologien-Migration*. Wiesbaden: Springer Vieweg, 1–648.
- [6] Bartosiak, Dave., 2017. "Exoskeletons unveiled at Ford manufacturing plant". The Drive [Internet]. [cited 2019.05.20]. Available from: <http://www.thedrive.com/sheetmetal/15921/exoskeletons-unveiled-at-ford-manufacturing-plant>
- [7] Bonomi, F., Milito, R., Natarajan, P., & Zhu, J., 2014. Fog computing: A platform for internet of things and analytics. In *Big data and internet of things: A roadmap for smart environments*. Springer, Cham, 169–186.
- [8] Brolin, A., Thorvald, P., & Case, K., 2017. Experimental study of cognitive aspects affecting human performance in manual assembly. *Production & Manufacturing Research*, 5(1), 141–163.
- [9] Brolin, A., Case, K., & Thorvald, P., 2016. Cognitive aspects affecting human performance in manual assembly. In *Advances in Manufacturing Technology XXX: Proceedings of the 14th International Conference on Manufacturing Research, Incorporating the 31st National Conference on Manufacturing Research*, IOS Press, 3, 231. Research, September 6–8, 2016, Loughborough University, UK.
- [10] Burke, R., Mussomeli, A., Laaper, S., Hartigan, M., & Sniderman, B., 2017. The smart factory: Responsive, adaptive, connected manufacturing. *Deloitte Insights*, 31.
- [11] De Looze, M. P., Bosch, T., Krause, F., Stadler, K. S., & O'Sullivan, L. W., 2016. Exoskeletons for industrial application and their potential effects on physical work load. *Ergonomics*, 59(5), 671–681.
- [12] Drahoš, P., Kučera, E., Haffner, O., & Klimo, I., 2018. Trends in industrial communication and OPC UA. In *2018 Cybernetics & Informatics (K&I)*. IEEE, 1–5.
- [13] Ebrahimi, A., 2017. Stuttgart Exo-Jacket: An exoskeleton for industrial upper body applications. In *2017 10th International Conference on Human System Interactions (HSI)*. IEEE, 258–263.
- [14] European Foundation for the Improvement of Living and Working Conditions, 2015. First findings: Sixth European working conditions survey: Résumé. Publications Office.
- [15] Deep-Burn Project: Annual Report for 2009, Idaho National Laboratory, Sept. 2009.
- [16] Ferguson, S. A., Marras, W. S., Allread, W. G., Knapik, G. G., & Splittstoesser, R. E., 2012. Musculoskeletal disorder risk during automotive assembly: current vs. seated. *Applied ergonomics*, 43(4), 671–678.
- [17] Ferrer, B. R., Mohammed, W. M., Chen, E., & Lastra, J. L. M., 2017. Connecting web-based IoT devices to a cloud-based manufacturing platform. In *IECON 2017-43rd Annual Conference of the IEEE Industrial Electronics Society*. 8628–8633.
- [18] Garofalakis, M., Gehrke, J., & Rastogi, R., 2016. *Data Stream Management: Processing High-Speed Data Streams*. Springer.
- [19] Monostori, L., Kádár, B., Bauernhansl, T., Kondoh, S., Kumara, S., Reinhart, G., & Ueda, K., 2016. Cyber-physical systems in manufacturing. *Cirp Annals*, 65(2), 621–641.
- [20] Henning, K., 2013. Recommendations for implementing the strategic initiative INDUSTRIE 4.0: Securing the future of German manufacturing industry; final report of the Industrie 4.0 Working Group. Forschungsunion.
- [21] Hitch, J., 2018. "Exo strategy: Preventing pain points with exoskeletons". NED [Internet]. [cited 2019.05.20]. Available from: <https://www.newequipment.com/industry-trends/exo-strategy-preventing-pain-points-exoskeletons>
- [22] Hunkeler, U., Truong, H. L., and Stanford-Clark, A. 2008. MQTT-S—A publish/subscribe protocol for Wireless Sensor Networks. In *2008 3rd International Conference on Communication Systems Software and Middleware and Workshops (COMSWARE'08)*. IEEE, 791–798.
- [23] Landau, K., Rademacher, H., Meschke, H., Winter, G., Schaub, K., Grasmueck, M., & Schulze, J., 2008. Musculoskeletal disorders in assembly jobs in the automotive industry with special reference to age management aspects. *International Journal of Industrial Ergonomics*, 38(7–8), 561–576.
- [24] Lee, J., Bagheri, B., & Kao, H. A., 2015. A cyber-physical systems architecture for industry 4.0-based manufacturing systems. *Manufacturing letters*, 3, 18–23.
- [25] Profanter, S., Tekat, A., Dorofeev, K., Rickert, M., & Knoll, A., 2019. OPC UA versus ROS, DDS, and MQTT: Performance Evaluation of Industry 4.0 Protocols. In *Proceedings of the IEEE International Conference on Industrial Technology (ICIT)*.
- [26] Schel, D., Henkel, C., Stock, D., Meyer, O., Rauhöft, G., Einberger, P., & Seidelmann, J., 2018. Manufacturing service bus: an implementation. *Procedia CIRP*, 67, 179–184.
- [27] Schnieders, T. M., & Stone, R. T., 2017. Current Work in the Human-Machine Interface for Ergonomic Intervention with Exoskeletons. *International Journal of Robotics Applications and Technologies (IJRAT)*, 5(1), 1–19.
- [28] Sylla, N., Bonnet, V., Colledani, F., & Fraisse, P., 2014. Ergonomic contri-

- bution of ABLE exoskeleton in automotive industry. *International Journal of Industrial Ergonomics*, 44(4), 475-481.
- [29] Tsardoulis, Emmanouil, and Pericles Mitkas, 2017. "Robotic frameworks, architectures and middleware comparison." arXiv preprint arXiv:1711.06842.
- [30] Wesslén, J., 2018 *Exoskeletons Exploration: Research, development, and applicability of industrial exoskeletons in the automotive industry* [Internet] [Dissertation]. Available from: <http://urn.kb.se/resolve?urn=urn:nbn:se:hj:diva-40093>
- [31] Zhong, R. Y., Xu, X., Klotz, E., & Newman, S. T., 2017. Intelligent manufacturing in the context of industry 4.0: a review. *Engineering*, 3(5), 616-630.
- [32] Zikopoulos, P., & Eaton, C., 2011. *Understanding big data: Analytics for enterprise class hadoop and streaming data*. McGraw-Hill Osborne Media.